Title & Project Overview

- **Title:** Carbon on the Chain: AI-Powered Sustainability Auditor

- **Subtitle:** Model Context Protocol (MCP) Track Submission | By Divya, SJB Institute of Technology

- **The Problem:** Traditional carbon credit trading suffers from inefficiencies, fraud, and a heavy reliance on manual dashboard verification.

- **The Solution:** Transitioned a standard Web3 dApp into an autonomous, AI-driven ecosystem. We built a "Sustainability Auditor" Agent that seamlessly queries Ethereum smart contracts and verifies corporate emission data using natural language, replacing manual UI clicks with intelligent automation.

Architecture & Implementation (The "How")

•**The AI Agent (Google ADK):** * Powered by Gemini 3.1 Pro with a strict "Compliance Officer" persona.

- Handles the reasoning: analyzes user requests and decides when to fetch external blockchain data.

•**The Tool Bridge (MCP Server):** * A custom Python server built with `FastMCP` and `Web3.py`.

- Exposes secure tools (`get_organization_details`, `verify_emission_report`) that bridge the AI's reasoning directly to local Ethereum smart contracts.

•**Cloud Deployment:** * The entire agent and web interface are containerized and deployed serverless on **Google Cloud Run** for scalable, secure access

Outcomes & Real-World Impact

•**Decoupled & Standardized:** Successfully separated AI reasoning from blockchain execution using the Model Context Protocol, creating a highly modular architecture.

•**Frictionless Auditing:** Eliminated the need for complex, multi-step auditor dashboards. Auditing is now as simple as asking: *"Check GreenTech's emissions and verify them on-chain.“*

•**Future Scalability:** By establishing the MCP foundation, the agent is primed to connect with other real-world data sources (like the Google Maps MCP for physical location validation or BigQuery for historical emission analytics) with zero changes to the core smart contracts.